A PROJECT REPORT ON

Identifying the Causes of Traffic Congestion at an Urban Junction

A Data-Driven Analysis using R

Submitted by

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Academic Year 2026 – 2027

ABSTRACT

Traffic congestion at signalised junctions imposes heavy costs on commuters and cities. This project investigates why congestion occurs at a single urban junction using six months of hourly data (4,392 observations) analysed entirely in R. The raw data, which contained duplicates, missing values, impossible sensor readings and inconsistent labels, was cleaned, enriched with derived variables and exported to Excel. Exploratory analysis showed that congestion is confined to weekday peak windows (08–11 h and 17–20 h): 15% of hours were High or Severe, and the worst hour, 18:00, averaged 13.2 minutes of delay at 9 km/h. Logistic regression and a decision tree classified congested hours with 98.7% and 96.1% test accuracy, and a regression on log-delay ranked the drivers. Traffic volume approaching capacity (about 1,000–1,200 vehicles per hour) explained 46.7% of the variation in delay, followed by road works, heavy-vehicle share, signal timing, rain and illegal parking; nearby events, holidays and pedestrian volume were unimportant. Recommendations include adaptive peak-hour signal timing, off-peak road works, parking enforcement, rapid signal-fault response and time restrictions for heavy vehicles. The dataset is simulated; the code can be re-run unchanged on real sensor data.

Keywords: traffic congestion, junction analysis, R, data cleaning, exploratory data analysis, regression, decision tree.

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1. Introduction

Traffic congestion at signalised junctions costs commuters time and cities money. A city wants to know **when** and **why** congestion occurs at one particular junction (Junction J1 – Nashik Phata Chowk, Pimpri-Chinchwad) so that limited resources can be directed to the causes that matter most. The objectives of this project are: (1) to clean a junction-level traffic dataset; (2) to describe congestion patterns using exploratory analysis in R; (3) to measure and rank the factors that cause delay; and (4) to recommend practical interventions. All work was done in **R 4.3** (dplyr, tidyr, readr, lubridate, ggplot2, corrplot, rpart, openxlsx); code, data and outputs are in the accompanying GitHub repository.

2. Dataset and Methodology

2.1 Dataset

The dataset holds **4,452 raw records (4,392 unique hourly observations)** for 1 April – 30 September 2026 with 19 variables: timestamp, vehicle count, average speed, queue length, delay, signal cycle and green time, heavy-vehicle share, pedestrians, rainfall, weather, and 0/1 flags for road work, accident, signal fault, nearby event, illegal parking and holiday, plus a congestion level (Low / Moderate / High / Severe). An hour is called **congested** when it is High or Severe. **The data is simulated** (script 01_generate_data.R) to behave like an urban junction and to contain realistic errors for the cleaning exercise, so the numbers are illustrative; the pipeline can be re-run unchanged on real sensor data.

2.2 Methodology

The pipeline has four R scripts: 01 generate raw data, 02 clean and export to CSV/Excel, 03 exploratory analysis with 8 figures, 04 modelling. Models were evaluated on a random 70:30 train–test split (fixed seed): a logistic regression classifier, a linear regression on log-delay (for percentage effects and importance) and a depth-4 decision tree.

3. Data Cleaning and Preparation

3.1 Data Quality Issues and Treatment

The audit found missing values, duplicates, impossible sensor readings (negative or 999 km/h speeds, hourly counts above 9,000) and text units inside numeric columns. Missing values were filled with the median of the same hour-of-day and weekday/weekend group, because traffic varies strongly through the day. The result has 4,392 rows and 29 columns (including new variables period, green_ratio, is_congested) with zero missing cells, exported as CSV and as an Excel workbook.

Table 1: Cleaning log

Problem found	Treatment	Rows
Exact duplicate records	Removed	60
Text units in vehicle_count (“812 veh”)	Stripped, converted to numeric	16
Inconsistent weather labels	Standardised to Title Case	328
Impossible speed / count / queue values	Set to NA, then imputed	12 / 8 / 7
Missing speed, count, queue, heavy-vehicle %	Same hour and day-type median	82 / 63 / 67 / 45
Missing weather / congestion level	From rainfall / decision tree	40 / 30

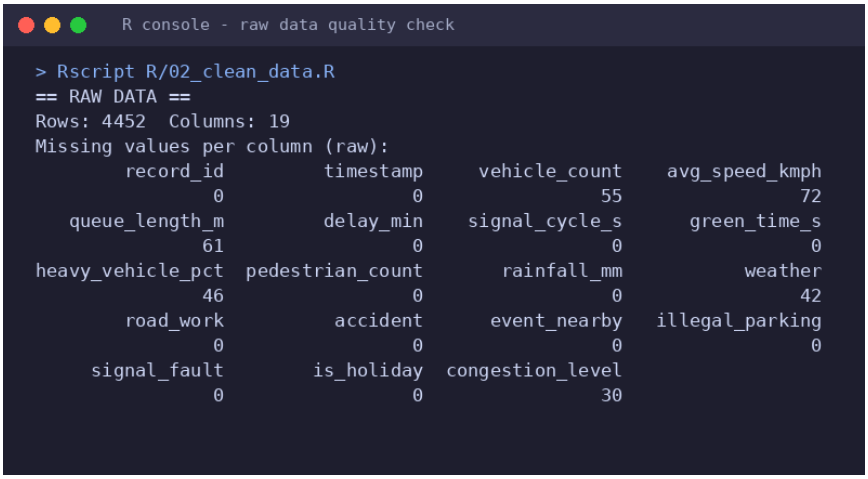


Figure 1: Raw data audit (R console)

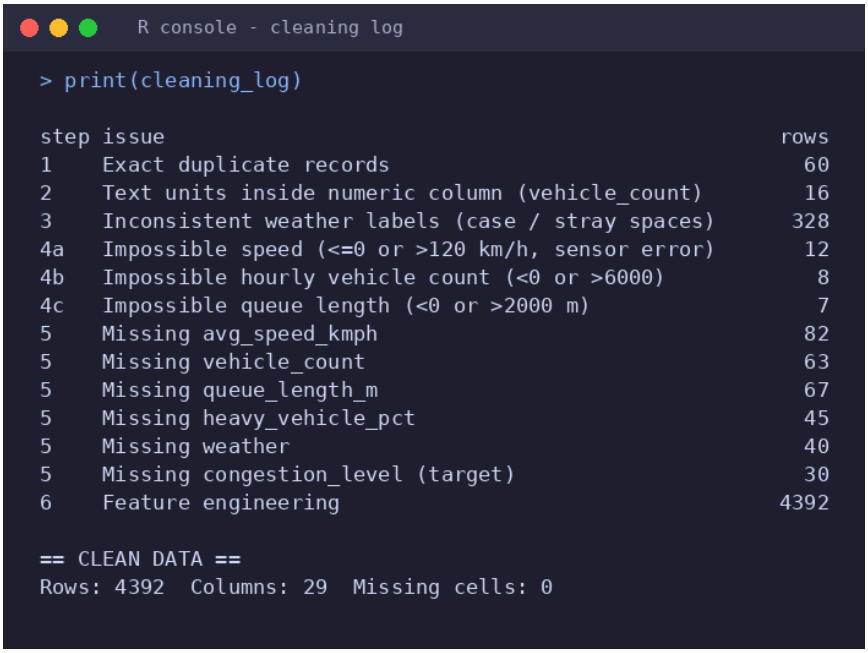


Figure 2: Cleaning log (R console)

4. Exploratory Data Analysis

4.1 When Does Congestion Occur?

Overall, 60.7% of hours were Low, 24.3% Moderate, 9.6% High and 5.4% Severe, so 15.0% (657 hours) were congested, but almost all of them fall in two weekday windows, 08–11 h and 17–20 h. The worst hour is 18:00, with an average delay of 13.2 minutes at 9.0 km/h. Weekends have a much smaller morning peak and almost no evening peak.

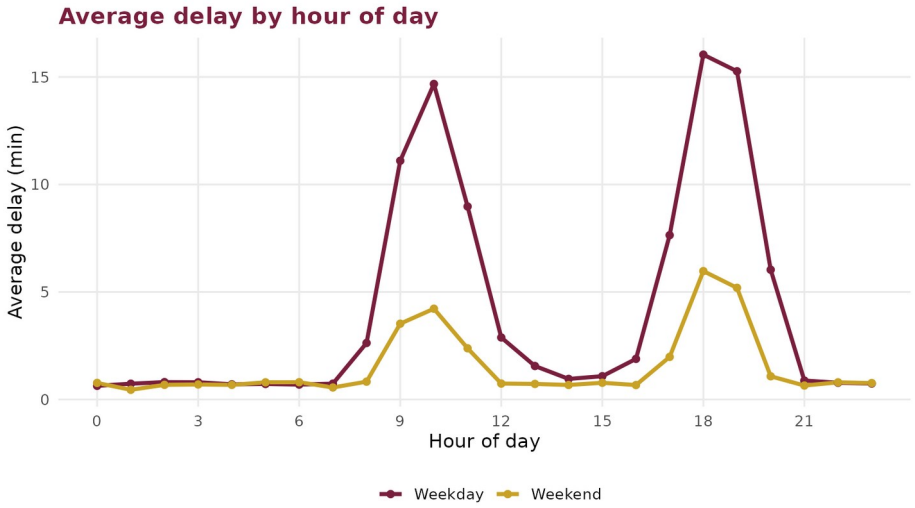


Figure 3: Average delay by hour, weekday vs weekend

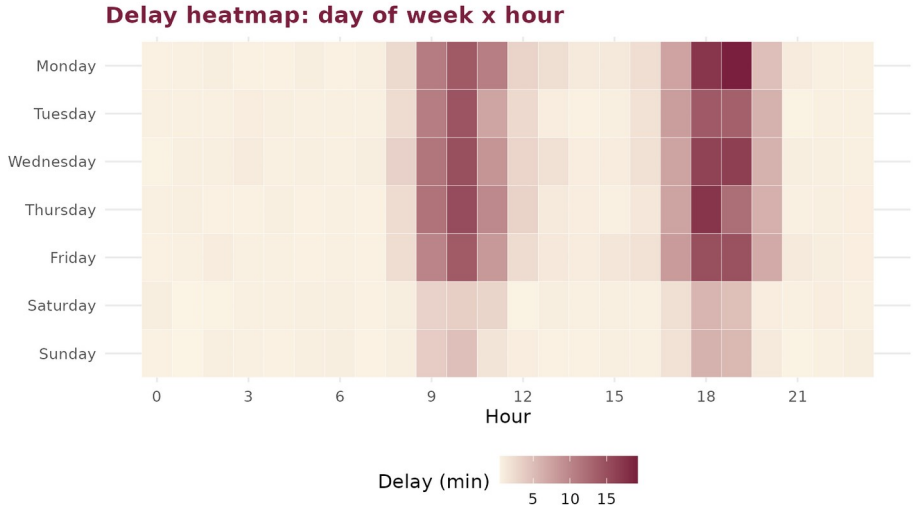


Figure 4: Delay heat map: day of week x hour

Table 2: Congestion by time period

Period	Hours	Vehicles/h	Speed (km/h)	Queue (m)	Delay (min)	% congested
Morning Peak (08–11)	732	1,273	15.3	320	7.5	39.3
Midday (12–16)	915	833	25.4	72	1.4	2.3
Evening Peak (17–20)	732	1,354	13.2	379	9.1	47.5
Off-Peak / Night	2,013	323	34.4	36	0.7	0.0

4.2 Volume and Capacity

Speed falls almost linearly as volume rises. Congestion is rare below 1,000 vehicles per hour (1.5% of hours at 800–1,000) but climbs to 10% at 1,000–1,200, 33% at 1,200–1,400 and 84.7% above 1,400. Delay correlates 0.98 with queue length, 0.67 with volume and −0.78 with speed.

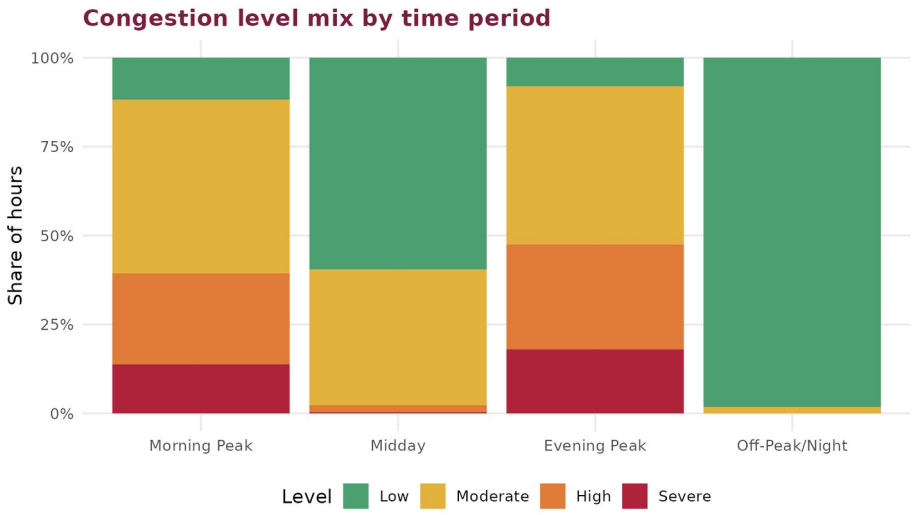


Figure 5: Congestion level mix by period

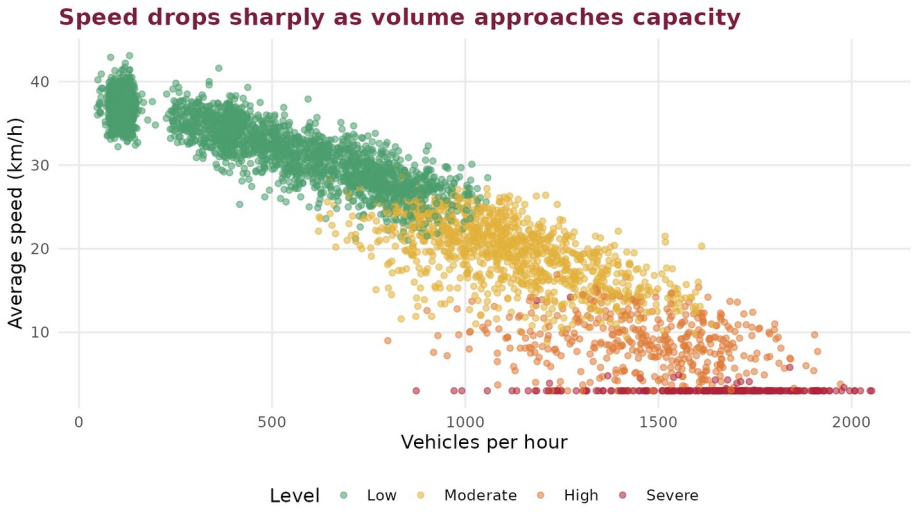


Figure 6: Speed versus traffic volume

4.3 Incidents and Conditions

Road works and signal faults raise the chance of congestion the most: 53.0% of hours are congested with road work (13.4% without) and 50.0% with a signal fault (14.7% without). Illegal parking and rain have a smaller effect but occur far more often (846 and 616 hours). Holidays and weekends show less congestion only because demand is lower. Congestion also rises from 12–13% of hours in April–May to 15–17% in the monsoon months.

Table 3: Congestion with and without each factor

Factor	% congested without	% congested with	Uplift (pp)	Delay without → with (min)	Hours present
Road work	13.4	53.0	+39.5	3.0 → 13.2	168
Signal fault	14.7	50.0	+35.3	3.3 → 12.1	34
Illegal parking	11.6	29.0	+17.3	2.8 → 5.9	846
Rain	13.4	24.7	+11.3	3.0 → 5.7	616
Accident	14.9	25.7	+10.8	3.4 → 6.3	35
Nearby event	14.9	16.3	+1.4	3.4 → 5.3	49

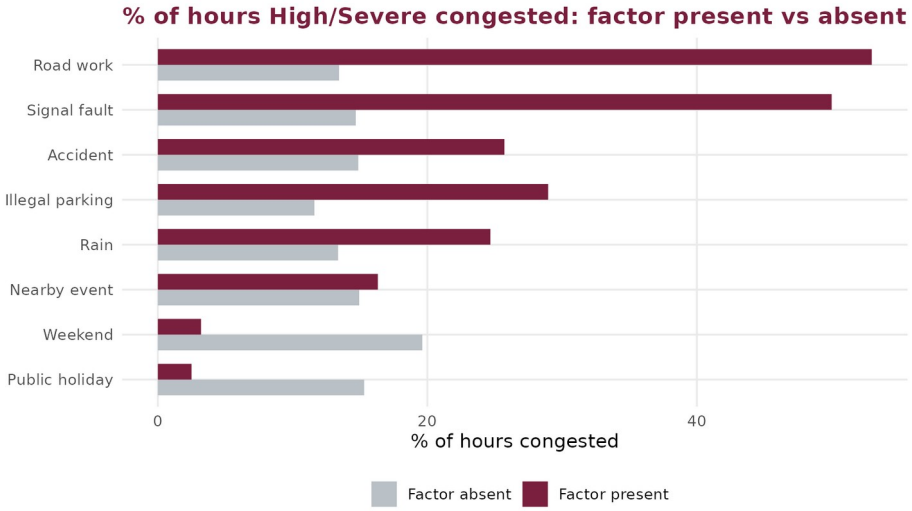


Figure 7: Congested hours with and without each factor

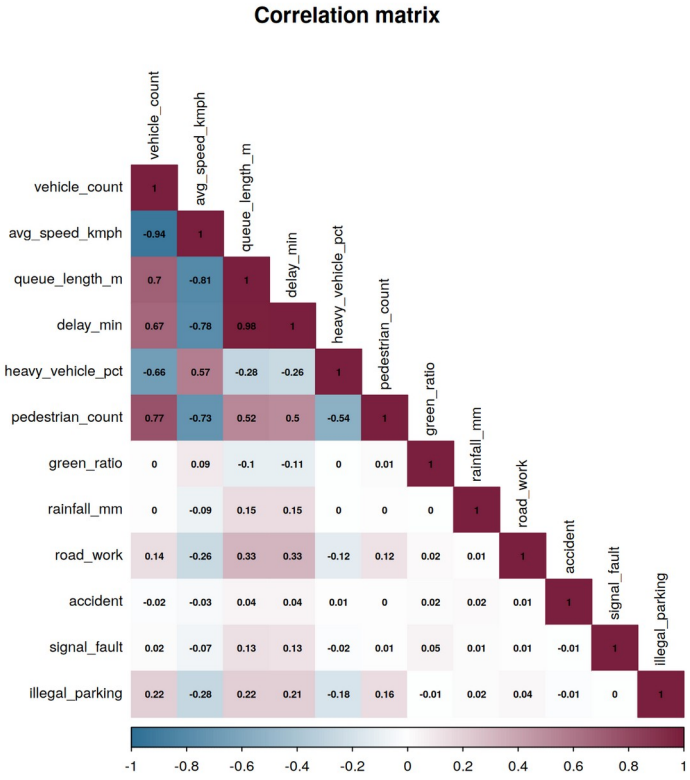


Figure 8: Correlation matrix

5. Modelling and Factor Importance

5.1 Model Accuracy and Effect of Each Factor

On the unseen 30% test data the logistic regression classified congested hours with **98.7% accuracy** (precision 97.0%, recall 94.6%, F1 95.8%, AUC 0.998) and the decision tree with 96.1%, showing that the recorded variables explain almost everything about congestion. A regression on log-delay (test $R^2 = 0.65$) gives the effect of each factor holding the others constant: **road work +226%, accident +154%, signal fault +117%, rain +52%, illegal parking +36%**, heavy-vehicle share +34% per 7 percentage points, and a higher green-time

ratio −14% per standard deviation. Nearby events (+22%, p = 0.16) and holidays (+2%, p = 0.85) are not significant.

5.2 Factor Importance

Importance is the share of explained variation in delay lost when a factor is removed: traffic volume 46.7%, time period 13.9% (mostly a proxy for demand), road work 12.1%, heavy-vehicle share 9.6%, green-time ratio 5.3%, rain 5.2%, illegal parking 3.2%, accident 1.8%, signal fault 1.2%; pedestrians, weekend, events and holidays are all below 1%. The decision tree splits first at about 1,364 vehicles per hour; between about 1,160 and 1,360, rain, road work, illegal parking and low green ratio decide whether a queue builds.

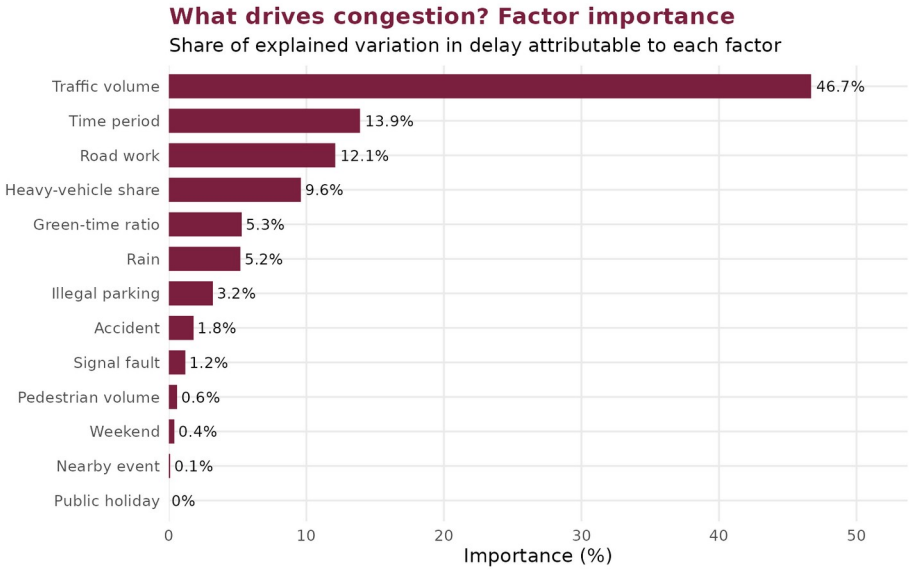


Figure 9: Factor importance for delay

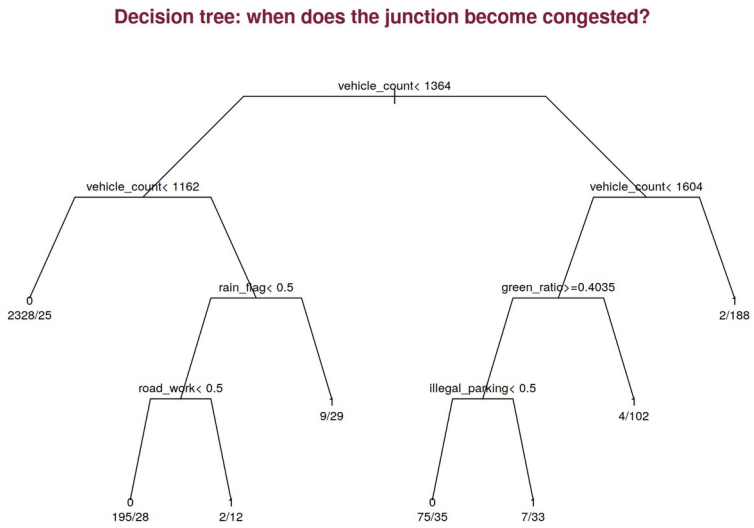


Figure 10: Decision tree for congested hours

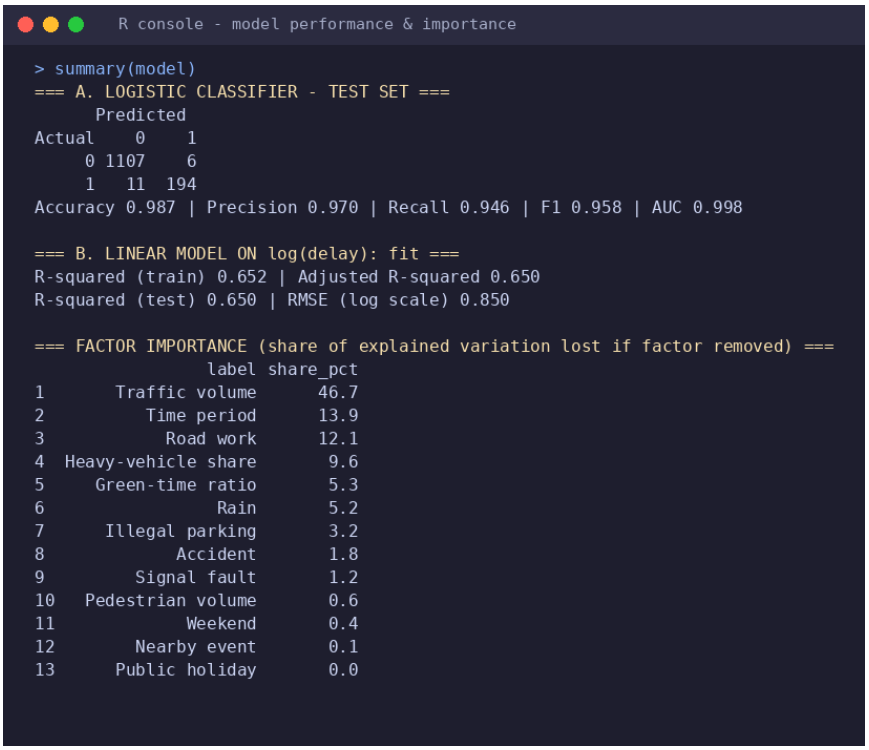


Figure 11: Model results (R console)

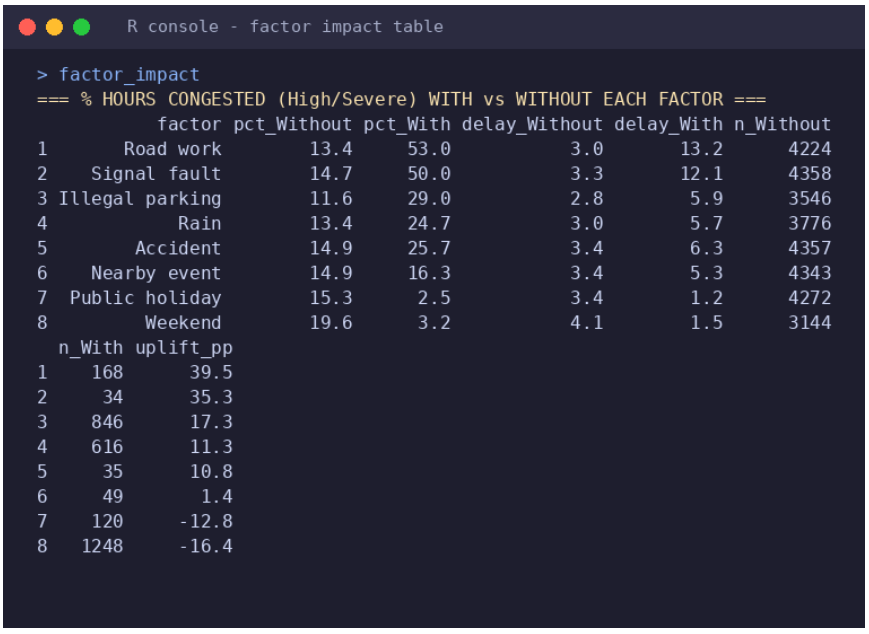


Figure 12: Factor impact table (R console)

6. Key Insights: What Matters and What Does Not

6.1 Important Insights

1. **Congestion is a weekday peak-hour problem:** 15% of hours, almost all in 08–11 h and 17–20 h; worst at 18:00.
2. **Demand versus capacity is the root cause:** volume explains 46.7% of delay; trouble starts at about 1,000–1,200 vehicles per hour.
3. **Road works are the most damaging factor** (+226% delay) and are fully schedulable.

4. **Illegal parking is the most frequent removable cause** (846 hours, +36% delay); **signal faults** are rare (34 hours) but half of them were congested.
5. **Rain (+52%)**, **heavy vehicles (+34% per 7 points)** and **signal timing (−14% per SD of green ratio)** matter at the margin.

6.2 Factors of Little Importance

Nearby events, public holidays, weekends and pedestrian volume have no independent effect once volume is known. Accidents are severe per event (+154%) but occurred in only 35 hours, so their overall share is small (1.8%).

7. Recommendations

1. Use adaptive peak-hour signal plans with longer green for the main approach during 08–11 h and 17–20 h.
2. Allow road works and lane closures only off-peak (about 21:00–06:00).
3. Enforce no-parking zones on the approaches with cameras and towing during 10–20 h.
4. Set up fast signal-fault response: alarms, spare controllers, battery back-up, repair within one hour.
5. Restrict heavy goods vehicles during the two peak windows and prepare a rain-day signal and deployment plan.
6. For the long term, consider a slip lane, grade separator or demand management, since evening volumes average about 1,350 and often exceed 1,400 vehicles per hour.

8. Conclusion and Limitations

8.1 Conclusion

Congestion at the studied junction is concentrated in two weekday peaks, driven mainly by demand approaching a capacity of about 1,000–1,200 vehicles per hour, and worsened by a few controllable factors: road works, signal faults, illegal parking and signal timing. Low-cost measures aimed at these factors should reduce delay before expensive infrastructure is considered.

8.2 Limitations and Future Work

The data is simulated, so percentages should not be quoted as real-world facts; only one junction and six months are covered; the models show association rather than proof of causation; and the log-delay model explains about 65% of the variation. Future work: real ATCS data, multiple junctions and a forecasting model for proactive signal control.

References: R Core Team, R: A Language and Environment for Statistical Computing; H. Wickham et al., "Welcome to the tidyverse", JOSS 4(43), 2019; T. Therneau and B. Atkinson, rpart; Transportation Research Board, Highway Capacity Manual. **To reproduce:** run Rscript R/01_generate_data.R, 02_clean_data.R, 03_eda.R and 04_model.R from the repository root.